

Problem when using ADAM optimizer with QAOA

<https://github.com/qiskit-community/qiskit-aqua/issues/1007>

ROW 111

Problem when using ADAM optimizer with QAOA #1007



Closed

#1016



EliasCombarro opened on May 21, 2020 · edited by Cryoris

Edits ...

Information

- **Qiskit Aqua version:** 0.7.1
- **Python version:** The one in the IBM Quantum Experience Jupyter notebook server
- **Operating system:** Program executed on the IBM Quantum Experience Jupyter notebook server

What is the current behavior?

When using ADAM as the optimizer for QAOA, an error "UnboundLocalError: local variable 'params_new' referenced before assignment" is raised after a few executions.

Steps to reproduce the problem

Running this program (I have simplified here the operator to minimize, but this happens also with other, more complicated operators and with the qasm_simulator backend)

```
from qiskit import BasicAer
from qiskit.aqua import aqua_globals, QuantumInstance
from qiskit.aqua.algorithms import QAOA
from qiskit.aqua.components.optimizers import ADAM
from qiskit.quantum_info import Pauli
from qiskit.aqua.operators import WeightedPauliOperator
pauli_list = [ [-1,Pauli([1],[0]]) ]
q_op = WeightedPauliOperator(paulis = pauli_list)
rep = 10
optimizer = ADAM()
backend = BasicAer.get_backend('statevector_simulator')
quantum_instance = QuantumInstance(backend)
qaoa = QAOA(q_op, optimizer, p=1)

for i in range(rep):
    result = qaoa.run(quantum_instance)
    #print(result)
    print("***** Iteration ", i, "*****")
    print("Optimal value", result['optimal_value'])
```

gives me the following output:

```
***** Iteration 0 *****
Optimal value -0.99999999997953
***** Iteration 1 *****
Optimal value -0.999999999343336

-----
UnboundLocalError                                Traceback (most recent call last)
<ipython-input-3-a86b02bdb77b> in <module>
      6
      7 for i in range(rep):
----> 8     result = qaoa.run(quantum_instance)
      9     #print(result)
     10     print("***** Iteration ", i, "*****")

/opt/conda/lib/python3.7/site-packages/qiskit/aqua/algorithms/quantum_algorithm.py in run(self, quantum_instar
     68         self.quantum_instance = quantum_instance
     69
--> 70         return self._run()
     71
     72     @abstractmethod

/opt/conda/lib/python3.7/site-packages/qiskit/aqua/algorithms/minimum_eigen_solvers/vqe.py in _run(self)
    348         var_form=self.var_form,
    349         cost_fn=self._energy_evaluation,
--> 350         optimizer=self.optimizer)
    351
    352     # TODO remove all former dictionary logic

/opt/conda/lib/python3.7/site-packages/qiskit/aqua/algorithms/vq_algorithm.py in find_minimum(self, initial_pc
    199         variable_bounds=bounds,
    200         initial_point=initial_point,
--> 201         gradient_function=gradient_fn)
    202
    203     eval_time = time.time() - start
```

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

[Fix ADAM reuseability](#)
qiskit-community/qiskit-aqua

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```

--> 201                                     gradient_function=gradient_fn)
202         eval_time = time.time() - start
203

/opt/conda/lib/python3.7/site-packages/qiskit/aqua/components/optimizers/adam_amsgrad.py in optimize(self, num
225                                     (objective_function, self._eps))
226
--> 227         point, value, nfev = self.minimize(objective_function, initial_point, gradient_function)
228         return point, value, nfev

/opt/conda/lib/python3.7/site-packages/qiskit/aqua/components/optimizers/adam_amsgrad.py in minimize(self, obj
194         params = params_new
195
--> 196         return params_new, objective_function(params_new), self._t
197
198     def optimize(self, num_vars, objective_function, gradient_function=None, variable_bounds=None,

UnboundLocalError: local variable 'params_new' referenced before assignment

```

Sometimes it takes several iterations, but eventually the error occurs.

What is the expected behavior?

The code should run without errors. When using other optimizer (COBYLA, for instance), the error does not happen.

Suggested solutions

The problem seems to be with the ADAM optimizer, seems replacing the optimizer solves the error.



woodsp-ibm on May 21, 2020 · edited by woodsp-ibm

Edits · Member · ...

Thanks for bringing this to our attention. The issue arises here in the ADAM optimizer, as its `optimize()` method is called each time `qaoa` is `run()` but the initialization of certain fields it uses during the optimization is only done in its constructor. It ends up working for a while before it gets into a state where it fails as you saw. For it to be able to be re-used in the way you are doing then things may need to be reset each time on the call to `optimize`. As we are moving towards allowing algorithms and components to be re-used this is something we will have to address.

Until such time as this is done you can re-create `QAOA` with `ADAM` each time you want to use it, or continue what you are doing but using the other optimizers. (I would have suggested using `qaoa.optimizer = ADAM()` to give it a new instance each time as an alternative. We just added setters in the last release but it seems there is an issue there too that I will have to fix! - Update: This is now fixed here on master branch by [#1008](#))

[@Zoufalc](#) - is fixing `ADAM` to allow its re-use across `optimize` something you can take a look at?

[@stefan-woerner](#) FYI. And I think the new `GSLS` is ok, from looking at it, in regard of re-use.



EliasCombarro on May 22, 2020

Author · ...

Thanks for the workaround. With that modification, my code seems to be running correctly now.

Thanks!

[Cryoris](#) added this to To Do in [Optimization](#) on May 25, 2020

[Cryoris](#) mentioned this on May 26, 2020

[Fix ADAM reuseability #1016](#)

[manoelmarques](#) closed this as [completed](#) in [#1016](#) on May 26, 2020

[manoelmarques](#) moved this from To Do to Done in [Optimization](#) on May 26, 2020

Fix ADAM resuseability #1016

Merged manaelmarques merged 3 commits into qiskit-community:master from Cryoris:adam-reusability on May 26, 2020

Conversation 0 Commits 3 Checks 0 Files changed 1

+62 -45



Cryoris commented on May 26, 2020

Contributor

Summary

Fixes #1007 (lines 181 and 191) and updates the documentation and type hints of ADAM .

Details and comments

The number of iterations was set in the initializer only, so re-running the optimization kept increasing the number of iterations. Since there's a maximum number of iterations, ADAM stopped the optimization.

Reviewers

- Zoufalc ✓
- woodsp-ibm ✓
- manaelmarques ●

Assignees

No one assigned

Labels

Changelog: Bugfix

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

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4 participants



Cryoris added 2 commits 5 years ago

reset step count in minimize and set params_new

Unverified

582f07b

remove duplicate loc

Unverified

95a889c

Cryoris requested review from manaelmarques and woodsp-ibm as code owners 5 years ago

Merge branch 'master' into adam-reusability

Verified

9b81586

Cryoris requested a review from Zoufalc 5 years ago

Zoufalc approved these changes on May 26, 2020

View reviewed changes

woodsp-ibm added the Changelog: Bugfix label on May 26, 2020

woodsp-ibm approved these changes on May 26, 2020

View reviewed changes

manaelmarques merged commit 35e6381 into qiskit-community:master on May 26, 2020

mtreinish pushed a commit to mtreinish/qiskit-core that referenced this pull request on Nov 20, 2020

Fix ADAM resuseability (qiskit-community/qiskit-aqua#1016)

Unverified

a9e0aacc

manaelmarques pushed a commit to manaelmarques/qiskit-terra that referenced this pull request on Dec 7, 2020

Fix ADAM resuseability (qiskit-community/qiskit-aqua#1016)

Unverified

19d13a7